

1) Contrast modification

```
I = imread("IMG_002.jpg");
gs = im2gray(I);
imshow(gs)
gsAdj = imadjust(gs);
imshowpair(gs,gsAdj,"montage")
imhist(gs)
imhist(gsAdj)
```

2) Thresholding

```
img = imread("IMG_004.jpg");
gs = im2gray(img);
imshow(gs)
```

```
BW = gs > 255/2;
imshow(BW)
```

```
imhist(gs)
```

```
BW = gs > 200;
imshow(BW)
```

3) Projection

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
```

```
BW = gs > 200;
imshowpair(I,BW,"montage")
```

```
S1 = sum(BW,1);
plot(S1)
```

```
S2 = sum(BW,2);
plot(S2)
```

```
BW = imbinarize(gsAdj,"adaptive","ForegroundPolarity","dark");
imshowpair(I,BW,"montage")
```

```
I2 = imread("IMG_005.jpg");
gs2 = im2gray(I2);
gs2Adj = imadjust(gs2);
BW2 = imbinarize(gs2Adj);
imshowpair(I2,BW2,"montage")
```

Task 1

```
S = sum(BW,2);
```

Task 2

```
plot(S)
```

Task 3

```
S2 = sum(BW2,2);
plot(S2)
```

4) Filter 1

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
BW = imbinarize(gs,"adaptive","ForegroundPolarity","dark");
imshowpair(gs,BW,"montage")

H = fspecial("average",3)
gssmooth = imfilter(gs, H);
imshow(gssmooth)
imshowpair(gs,gssmooth,"montage")

BWsmooth = imbinarize(gssmooth,"adaptive","ForegroundPolarity","dark");
imshow(BWsmooth)
```

Task 4

```
gssmooth = imfilter(gs,H,"replicate");
```

Task 5

```
BWsmooth = imbinarize(gssmooth,"adaptive","ForegroundPolarity","dark");
imshow(BWsmooth)
```

5) Filter 2

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
H = fspecial("average",3);
gs = imfilter(gs,H,"replicate");
imshow(gs)
```

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
H = fspecial("sobel")
gs = imfilter(gs,H,"replicate");
imshow(gs)
```

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
H = [1 1 1; 0 0 0; -1 -1 -1]
gs = imfilter(gs,H,"replicate");
imshow(gs)
```

```
I = imread("IMG_004.jpg");
gs = im2gray(I);
H = [1 0 -1; 1 0 -1; 1 0 -1]
gs = imfilter(gs,H,"replicate");
imshow(gs)
```

```
SEdisk = strel("disk",8);
Ibg = imclose(gs,SEdisk);
gsSub = Ibg - gs;
BW = ~imbinarize(gsSub);
imshowpair(I,BW,"montage")
```

Task 1

```
SE = strel("rectangle",[3 25]);
```

Task 2

```
BWstripes = imopen(BW,SE);
imshow(BWstripes)
```

Task 3

```
S = sum(BWstripes,2);
plot(S)
```

Task 4

```
Sbw = sum(BW,2);
hold on
plot(Sbw)
hold off
```